Was the quantum ever load-bearing?

Auditing the classical simulability of trained quantum-NLP text models

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Abstract

Quantum natural language processing (QNLP) models—compositional DisCoCat circuits, quantum self-attention networks, and all-quantum token mixers—are promoted for a quantum benefit on text classification. That benefit is usually argued from accuracy, yet whether a *trained* QNLP circuit ever leaves the efficiently classically-simulable regime is rarely tested. We introduce SimCert, an open, model-agnostic protocol that audits this directly: it re-simulates each trained circuit as a bond-dimension-χ matrix product state, sweeps χ, and records the smallest χ (denoted $\chi^\star$) at which a classical surrogate recovers the model's own predictions, alongside an entanglement-removal ablation and two dequantization witnesses. Auditing five published QNLP architectures across four datasets (up to 10 qubits), we find that a classical surrogate recovers the models' predictions at *small, non-growing* bond dimension: attention-style models are recovered at χ=1 (a product state), and a compositional model at χ=2; $\chi^\star$ stays flat as we scale the qubit count while the exact bound grows exponentially. The circuits do carry entanglement, but removing it barely moves the result. We read this cautiously—it *suggests* the quantum resource is not the operative ingredient *at the scales these models are tested*, not that QNLP is classically simulable in general. We release the harness and phrase the findings as a protocol and a set of questions for QNLP model and benchmark design.

1 Introduction

Because fault-tolerant quantum hardware is not yet available, most claims in quantum machine learning (QML) are established by classical simulation of small circuits, and quantum NLP inherits this practice. A steady stream of QNLP models reports accuracy gains on text classification: compositional DisCoCat pipelines run on and simulated for real devices [Lorenz et al., 2021], quantum self-attention networks [Li et al., 2022, Chen et al., 2024], and, more recently, fully-quantum token mixers [Chen et al., 2025]. The reported numbers are real. What is usually left implicit is the inference the reader is invited to draw—that the *quantum* structure is doing the work.

That inference is testable, and it is tested surprisingly rarely. Across the QNLP text-classification papers we examined (Appendix A), models are compared against classical *networks* of matched size but not against a classical *simulation of their own circuit*—the one baseline that would isolate whether the quantum resource, rather than the surrounding classical machinery or the trainable parameters, is responsible.[1] The same gap was recently closed for tabular QML [Bowles et al., 2024] and for quantum convolutional networks [Bermejo et al., 2024], with the theoretical footing that trainable variational models are constrained matrix product states (MPS) [Shin et al., 2024]

[1] We count a paper as testing simulability only if it reports a classical surrogate of the trained circuit (tensor-network contraction, Pauli propagation, or an explicit entanglement-removal ablation), not merely a separately-trained classical baseline. This is a focused review, not a saturated systematic survey; scope and counts are in Appendix A.

and that the absence of barren plateaus tends to imply classical simulability [Cerezo et al., 2025]. *So which QNLP results survive a classical surrogate of the trained circuit?* That is the question we take up.

We make three contributions, all of them audit verbs rather than claims of a new model.

- We **define** a scoped simulability criterion for trained QNLP circuits—the smallest MPS bond dimension $\chi^\star$ at which a classical surrogate recovers the model's predictions—and package it with an entanglement-removal ablation, entanglement entropy, a geometric-difference kernel test, and a data-reuploading spectral test into a per-model *simulability certificate*.
- We **audit** five published QNLP architectures (DisCoCat/`lambeq`, quantum self-attention `QSANN` and `QMSAN`, the all-quantum token mixer `CLAQS`, and a variational-classifier reference) across four datasets, and report a $\chi^\star$-vs-size scaling study.
- We **release** an open, model-agnostic harness that consumes any trained circuit through a common intermediate representation, so the certificate can be recomputed and extended by others.[2]

The paper front-loads its epistemics. Section 2 sets out what a simulability audit can and cannot establish; Section 3 fixes definitions and the certificate; Sections 4–5 describe the audited models, data, and surrogate; Section 6 reports the findings; Section 7 states the caveats and the open questions we think follow.

2 What a simulability audit can and cannot tell us

A skeptical result earns trust only if it anticipates the ways it could be wrong, so we state the limits before the evidence. An audit of this kind speaks to *trained* models at the *scales they are actually run*. It does not, and cannot, prove that no QNLP model at any size resists classical simulation; the interesting asymptotic question is exactly the one that simulation cannot reach. What it can do is check a concrete, load-bearing assumption: that the accuracy a model reports depends on quantum structure that a cheap classical surrogate cannot reproduce.

Three failure modes shape the design. First, a surrogate that is too weak will fail to recover predictions for uninteresting numerical reasons; we therefore sweep the surrogate's power (the bond dimension) rather than fixing it, and report the whole curve. Second, "recovering accuracy" and "recovering the model's function" are different tests—a surrogate could match the label while computing a different function—so we report agreement with the trained model's own predictions, not only accuracy against ground truth. Third, absence of a quantum effect on a saturated benchmark says little; we therefore include a genuinely hard task and are explicit about where a benchmark is degenerate. None of this settles the asymptotic question. It settles the empirical one that current QNLP papers implicitly answer in the affirmative.

3 Framework and definitions

Setup. A QNLP text classifier maps an input x (a sentence, or a sequence of token embeddings) to a trained circuit that prepares a state $|\psi_\theta(x)\rangle$ on n qubits, from which a readout—a set of Pauli expectation values, possibly followed by a small classical head—produces a label $\hat{y}_\theta(x)$. We treat the trained parameters θ as fixed and ask about the function $\hat{y}_\theta$.

[2] Anonymized for review: `https://anonymous.4open.science/r/simcert`.

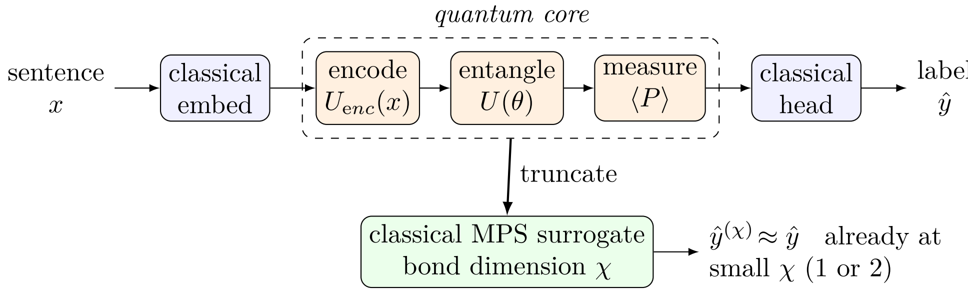

Figure 1: **What the audit interrogates.** A trained QNLP classifier (top) threads a sentence through a classical embedding, a *quantum core* (encode → entangle → measure), and a classical head. SimCert replaces the quantum core with a classical matrix-product-state surrogate of bond dimension χ (bottom) and asks how small χ can be before the model's predictions change. For every audited model the answer is small and constant: the surrogate reproduces the predictions at χ=1 (a product state) or χ=2.

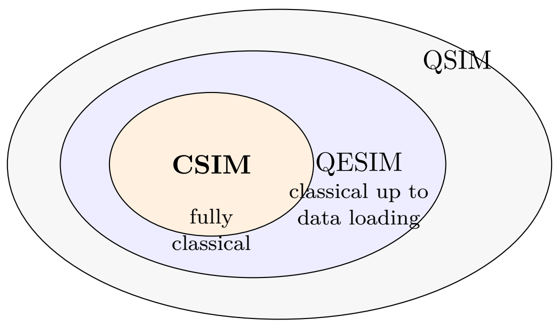

Figure 2: **Which simulability we claim.** The standard hierarchy: classically simulable (CSIM) $\subseteq$ classically simulable up to a data-loading step (QESIM) $\subseteq$ quantum-simulable (QSIM) [Cerezo et al., 2025, Shin et al., 2024]. Our MPS surrogate realises CSIM when the required bond dimension is small; we make no claim outside that regime.

Which notion of simulability. We adopt the standard hierarchy: a model is *classically simulable* (CSIM) if its outputs can be computed classically in polynomial time, and *efficiently classically simulable up to a data-loading step* (a weaker, common-in-practice notion) if the only obstruction is preparing the input state [Cerezo et al., 2025, Shin et al., 2024]. Our surrogate is an MPS, which realizes CSIM whenever the required bond dimension is small; we make no claim about regimes where it is not. Every claim below is of the form "simulable at bond dimension up to χ, at these model sizes, on these datasets."

The bond-dimension criterion. For a trained state we form its optimal bond-dimension-χ MPS approximation $|\psi^{(\chi)}\rangle$ by sequential singular-value truncation, recompute the readout, and obtain a truncated prediction $\hat{y}^{(\chi)}(x)$. We report, over the test set, the truncated accuracy $A(\chi)$, the agreement $\pi(\chi) = \Pr_x[\hat{y}^{(\chi)}(x) = \hat{y}(x)]$ with the untruncated model, and the mean state fidelity $F(\chi)$. The certificate's headline scalar is

$$\chi^\star \;=\; \min\{\chi : A_{\text{full}} - A(\chi) \leq \tau\}, \tag{1}$$

the smallest bond dimension whose accuracy is within a tolerance τ of the untruncated model. We set τ to the model's own generalization gap (differences smaller than the gap the model already

Model	Paradigm	Cross-token mixing	Readout	State
`discocat`	compositional	grammar cups (post-selection)	open-wire Born rule	pure
`qsann`	attention	Gaussian of measured Q/K (classical)	Pauli vector → head	pure
`qmsan`	attention	$\mathrm{tr}(\rho_q \sigma_k)$ (mixed-state)	Pauli vector → head	mixed
`claqs`	token mixer	complex LCU + QSVT polynomial	XYZ Paulis → MLP	pure
`vqc_text`	reference	entangling ring	$\langle Z_0 \rangle$	pure

Table 1: **The audited models.** One row per architecture, grouped by paradigm. `qmsan` is the only mixed-state model and is audited through the purification of its reduced density matrices. Full circuits, gate sets, and parameter counts are in Appendix B.

exhibits are not meaningfully "quantum resource") and report robustness to two alternative tolerances in the appendix. A model whose $\chi^\star$ is small *and does not grow* with n is, in the operational sense above, recovered by a cheap classical surrogate; we call such a model "bond-cheap" and use the term bare thereafter.

Result (informal). *For every audited model at the tested scales, $\chi^\star$ is a small constant (one or two) and does not grow when the qubit count is increased on a fixed task, while the bond dimension required for an* exact *MPS grows as $2^{\lfloor n/2 \rfloor}$.*

Corroborating witnesses. Bond dimension is one lens. The certificate adds three inexpensive ones: (i) an *entanglement-removal* ablation that retrains a product-state twin of the model and reports $\Delta A_{\text{ent}} = A_{\text{full}} - A_{\text{sep}}$, after Bowles et al. [2024]; (ii) the mean bipartite entanglement entropy $\bar{S}$ of the trained states, which lower-bounds the χ an exact representation needs; and (iii) two dequantization probes where they apply—the geometric difference g_{CQ} between the trained quantum kernel and a classical kernel [Huang et al., 2021] and the effective data-reuploading Fourier degree [Schuld et al., 2021]. We report the full vector rather than collapsing to a single verdict, because, as Section 6 shows, the witnesses can legitimately disagree.

4 Models and datasets audited

Selection. We audit published QNLP text classifiers that (a) target sentence/text classification, (b) fit within a simulator's reach at their published qubit counts, and (c) can be reimplemented from their descriptions. Applying these criteria to the QNLP literature leaves a compact set spanning the two dominant paradigms—compositional and attention-based—plus a recent all-quantum mixer; we add one plain variational classifier as a reference point. Table 1 compares them at a glance and stands in for a wall of per-model circuit diagrams (representative ansätze are drawn in Appendix B). None of the attention/mixer models ship official code, so we reimplement from the equations and report the reproduction gap openly (Section 6); the compositional model uses the official `lambeq` pipeline.

Datasets. We use the tasks these models were built on. MC (meaning classification, cooking vs. computing) and RP (a relative-pronoun task) are the canonical DisCoCat benchmarks of Lorenz et al. [2021]; MC is small and near-separable, RP is the harder, low-data task where published accuracies actually differ across models. We add a balanced subset of SST-2 as a genuinely non-separable sentiment task, and a synthetic MC-style set for controlled scaling. Sizes and preprocessing are in Appendix C. Two honest limitations are worth stating here rather than burying: our RP

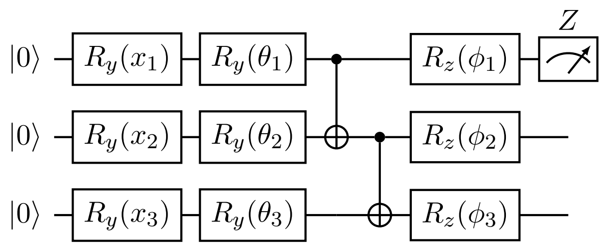

Figure 3: **Representative variational ansatz** (the reference classifier; the attention and mixer models use the same primitives). Each input feature is angle-encoded ($R_y(x_i)$), followed by a trainable layer ($R_y(\theta_i)$), a CNOT entangling chain, and a trainable $R_z(\phi_i)$. The block repeats L times with the data re-uploaded each layer, and the class is read from $\langle Z_0 \rangle$. The audit forms the bond-dimension-χ MPS approximation of the state prepared *just before* measurement. Per-model circuits (including `qmsan`'s mixed-state readout and `claqs`'s LCU/QSVT block) are in Appendix B.

Algorithm 1 SimCert bond-dimension audit for one trained model

input: trained model M, test set D, tolerance τ, bond dimensions X
for each example $x \in D$ **do**
    export the trained units of x (one whole-sentence circuit, or per-token circuits)
    for each $\chi \in X$ **do**
        truncate each unit's state to bond dimension χ; recompute its Pauli readout
        $\hat{y}^{(\chi)}(x) \leftarrow$ model's own decision rule applied to the truncated readout
    end for
end for
$A(\chi), \pi(\chi), F(\chi) \leftarrow$ accuracy, agreement, fidelity aggregated over D
$\chi^\star \leftarrow \min\{\chi : A_{\text{full}} - A(\chi) \leq \tau\}$ ▷ Eq. 1
return certificate ($\chi^\star,\ \Delta A_{\text{ent}},\ \bar{S},\ g_{CQ},$ Fourier degree)

reimplementations use our own token embeddings, and our DisCoCat pipeline uses an offline reader in place of the trained grammar parser whose model host is no longer reachable. Both cost accuracy on RP, as we report.

5 The classical surrogate and simulation protocol

We do not conjecture simulability; we realize it. Each trained model exports its per-example circuit to a common representation—OpenQASM for gate circuits, or, for the token mixer whose state is a matrix polynomial rather than a gate sequence, the trained statevector directly. A single audit consumer then re-simulates every model identically (Algorithm 1), so the audit never depends on which framework produced a circuit. The MPS truncation itself is an exact sequential-SVD computation, cross-checked against an independent tensor-network simulator; on circuits with known answers (a GHZ state, a product state) it reproduces the analytic bond structure, which is our correctness check for the whole pipeline.

The surrogate's cost is explicit: an MPS sweep on an n-qubit, depth-d circuit scales as $\mathcal{O}(n\,d\,\chi^2)$, and the exact-representation bound is $\chi = 2^{\lfloor n/2 \rfloor}$, so a small measured $\chi^\star$ against a rapidly growing exact bound is precisely the signal we are after. All experiments run on a single CPU (no GPU); we report seeds and, where a reimplementation is heavier than the hardware comfortably allows, we say so and reduce scale rather than quietly dropping runs. Full configurations are in Appendix D.

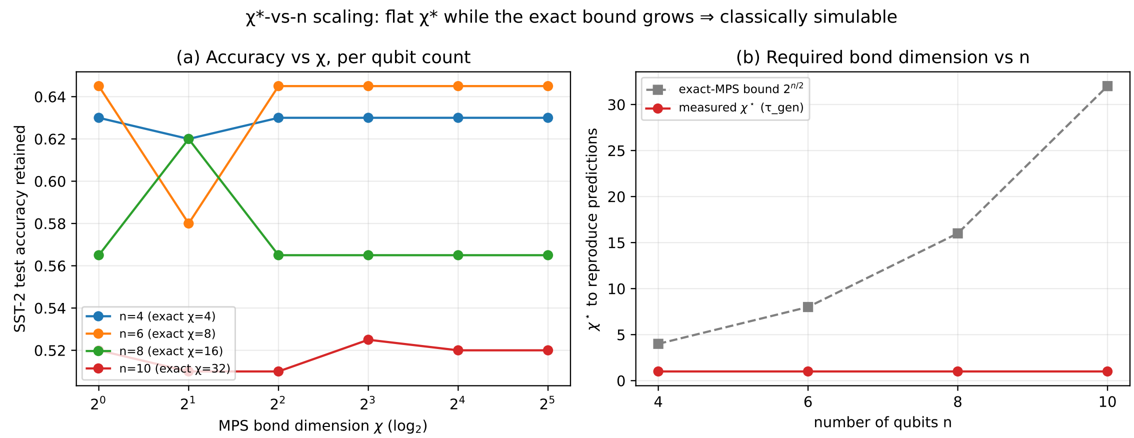

Figure 4: $\chi^\star$ **does not grow with qubit count.** (a) Accuracy retained vs. bond dimension χ for the reference model at $n \in \{4, 6, 8, 10\}$ on SST-2; the curves are flat down to χ=1. (b) Measured $\chi^\star$ (red) stays at 1 while the exact-MPS bound $2^{n/2}$ (grey, dashed) grows exponentially. A product-state surrogate recovers the predictions at every size tested. *Caveat:* accuracy here is modest and does not rise with n, so this probes recovery, not a strong classifier.

6 Results

We report three findings, then a subsection of effects we looked for and did not see. Unless noted, numbers are mean over three seeds; the aggregate table is Table 2 and the reproduction gap is Table 3.

6.1 A classical surrogate recovers the predictions at small bond dimension

Across models and datasets, a bond-dimension-one or -two surrogate recovers the trained model's predictions. On the canonical MC task, the attention and mixer models (`qsann`, `qmsan`, `claqs`) and the reference classifier are all recovered at χ=1—a *product state*—with no accuracy lost; the compositional `discocat` needs χ=2, but no more. The pattern holds on real data: our DisCoCat run reproduces the published MC accuracy to within 0.02 (Table 3) and is recovered at χ=2. The scales, in other words, are already low. In the operational sense of Section 3, these trained models are bond-cheap.

6.2 $\chi^\star$ does not grow with system size

The obvious objection is that small circuits are trivially simulable, so the audit only bites at scale. Figure 4 tests this directly: on a fixed task we increase the qubit count of the reference model from 4 to 10 and track $\chi^\star$. It stays flat at 1 while the exact bond bound grows $4 \to 8 \to 16 \to 32$. Adding qubits neither raises the bond dimension a classical surrogate needs nor—on this task—improves accuracy. *The gap between what the model uses and what it could use widens with every qubit.*

6.3 Removing the quantum resource barely moves the result

Two further witnesses agree. The trained circuits are not trivial product states—their mean bipartite entanglement entropy ranges from about 0.1 to 1.3 nats (Table 2)—yet truncating to χ=1 loses no predictions for the attention models, so the entanglement they build is not load-bearing

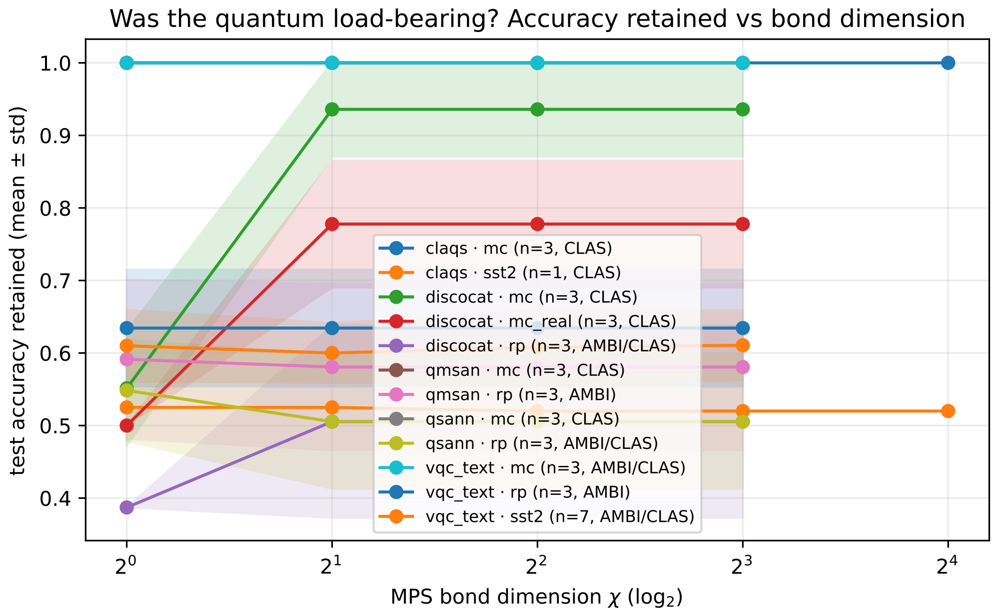

Figure 5: **Accuracy retained vs. bond dimension.** One curve per (model, dataset), mean±std over seeds. Flat curves down to χ=1 are the signature of a bond-cheap model: the classical surrogate loses nothing as it is weakened. SST-2 sits lower (a real, non-separable task) but is equally flat in χ.

for their decisions. The entanglement-removal ablation tells the same story on the easy tasks: a retrained product-state twin matches the full model ($\Delta A_{\text{ent}} \approx 0$). Figure 5 shows the accuracy-vs-χ curves; they are flat where the models saturate, and the one place the state fidelity climbs gradually with χ (the 8-qubit mixer) is also the one place the bond-dimension axis has room to resolve structure—and even there the *predictions* are recovered at χ=1.

6.4 Effects we did *not* observe

We looked for, and did not find, three things. We did *not* see $\chi^\star$ grow with qubit count on a fixed task (Figure 4). We did *not* see the entanglement-removal ablation cost accuracy on the tasks where the models saturate. And we did *not* see the compositional model behave like the attention models: `discocat` alone needs χ=2, and truncating it to χ=1 collapses its accuracy toward chance—its grammar cups create entanglement that *is*, minimally, load-bearing. This lone exception is worth more than the uniform cases: it shows the audit is not simply reporting "everything is χ=1" by construction.

Reproduction, honestly. Table 3 places our reimplementations against the published numbers. On MC the compositional model reproduces to within 0.02. On RP—a 74-example task the original papers themselves flag as fragile—our numbers run 0.17–0.22 below published, which we attribute to our own token embeddings and, for DisCoCat, the offline reader standing in for the unavailable grammar parser. We flag this plainly because it matters for one thing and not another: it weakens any claim about matching RP accuracy, but the simulability verdict is about our *own* trained circuits and is unaffected by whether they hit the published accuracy. The certificate audits the model in hand.

Model	Data	seeds	Full acc	Acc@χ=1	$\chi^\star$	$\bar{S}$ (nats)	ΔA_{ent}
`claqs`	mc	3	1.000±0.000	1.000±0.000	1	1.017±0.158	–
`claqs`	sst2	1	0.520	0.525	1	0.517	–
`discocat`	mc	3	0.936±0.065	0.551±0.079	2	0.659±0.013	–
`discocat`	mc_real	3	0.778±0.087	0.500±0.000	2	0.667±0.019	–
`discocat`	rp	3	0.505±0.133	0.387±0.000	1/2	0.654±0.005	–
`qmsan`	mc	3	1.000±0.000	1.000±0.000	1	0.169±0.077	0.000±0.000
`qmsan`	rp	3	0.581±0.115	0.591±0.110	1	0.106±0.022	-0.065±0.053
`qsann`	mc	3	1.000±0.000	1.000±0.000	1	0.295±0.064	0.000±0.000
`qsann`	rp	3	0.505±0.092	0.548±0.070	1	0.400±0.012	-0.022±0.030
`vqc_text`	mc	3	1.000±0.000	1.000±0.000	1	0.836±0.115	0.051±0.036
`vqc_text`	rp	3	0.634±0.080	0.634±0.080	1	0.685±0.059	0.097±0.121
`vqc_text`	sst2	7	0.610±0.050	0.610±0.050	1	1.282±0.232	0.123±0.033

Table 2: **Simulability certificate, all models.** Mean±std over seeds. *Acc*@χ=1 is the accuracy of a product-state surrogate; $\chi^\star$ is Eq. 1; $\bar{S}$ is mean bipartite entanglement entropy (nats); ΔA_{ent} is the entanglement-removal gap ("–" where a separable twin is not defined). Attention/mixer models are recovered at χ=1; `discocat` needs χ=2.

Model	Data	seeds	ours (test)	published
`discocat`	mc_real	3	0.778±0.087	0.798
`discocat`	rp	3	0.505±0.133	0.723
`qmsan`	rp	3	0.581±0.115	0.756
`qsann`	rp	3	0.505±0.092	0.677
`vqc_text`	sst2	7	0.610±0.050	–

Table 3: **Reproduction gap.** Our audited accuracy vs. published, on the datasets with a published number. MC_REAL reproduces to within 0.02; RP runs below published (fragile 74-example task; our embeddings and, for DisCoCat, an offline reader). The simulability verdict concerns our trained circuits and is independent of this gap.

7 Discussion, caveats, and open questions

The scoped claim is narrow and, we think, well supported: the QNLP models we audited reproduce their behavior under a classical surrogate of small, non-growing bond dimension, and the entanglement they carry is (with one revealing exception) not load-bearing for their predictions, *at the model sizes and datasets tested*. We are careful not to say more.

Several caveats bound the reading. The audit is average-case over a test set, not a statement about every input; $\chi^\star$ is defined against a tolerance, and while we check robustness to its choice, a different tolerance shifts the borderline cases. Our simulability notion is the practical CSIM-up-to-data-loading one, not a hardness result. The scales are small—the point of the scaling study is precisely that they need not be large for the pattern to appear, but a determined optimum at larger n remains out of simulation's reach. And the all-quantum mixer is run at reduced scale on a CPU; a faithful full-scale reproduction of its published accuracy would need a GPU, and is the one experiment here we could not run.[3]

There is a constructive reading. If these models are bond-cheap at deployment scale, they can be run classically today, which is useful in its own right; and the audit says exactly where

[3] This is also the only place additional compute would change what we can measure; the rest of the audit is CPU-bound by design.

a genuine quantum ingredient would have to live to matter—in a regime where $\chi^\star$ grows. That reframing turns the negative headline into a design target, and it suggests concrete questions for QNLP. Which architectural choices, if any, make $\chi^\star$ grow with n? Do the compositional cups that make `discocat` need χ=2 point toward structure worth scaling? Should QNLP benchmarks report a simulability certificate alongside accuracy, the way tabular QML now increasingly does? Our position is collegial rather than combative: the tabular and vision audits reached the same conclusion in their domains [Bowles et al., 2024, Bermejo et al., 2024], and we read our results as extending, not contradicting, theirs. *Until a QNLP model shows its bond dimension growing, the burden of proof for a quantum benefit sits with the model, not the skeptic.*

Reproducibility statement

The harness, all model reimplementations, exact configurations, seeds, dataset indices, and the scripts that regenerate every figure and table from stored results are released at the anonymized repository above. The audit is deterministic given a trained model; the only stochasticity is in training, which we control by seed.

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A Literature-review methodology

The motivating claim in Section 1—that QNLP text-classification papers argue a quantum benefit from accuracy but rarely test whether the *trained circuit* is classically simulable—rests on a focused review, not an exhaustive systematic survey; we scope it honestly here.

Search and inclusion. We searched arXiv (`quant-ph`, `cs.CL`) and Semantic Scholar for the conjunctions of {"quantum"} with {"natural language processing", "text classification", "self-attention", "sentence classification"} over 2019–2025, and followed citations from the models we reimplement. We include a paper if it (i) proposes a quantum or hybrid model for a text-classification task and (ii) reports an accuracy comparison. We record, for each, whether it reports a *classical surrogate of the trained circuit*—a tensor-network contraction, Pauli-propagation estimate, or an explicit entanglement-removal ablation—as opposed to merely a separately-trained classical network of comparable size.

What we found. Table 4 lists the core set. All report a classical *network* baseline; none report a classical surrogate of their own trained circuit, and none report the bond dimension, entanglement, or a simulability check of the trained model. This is the gap SimCert fills. We stress the limits: this is a focused review of the papers central to this study plus those surfaced by the searches above, not a saturated systematic survey, and we did not attempt inter-rater agreement. We therefore phrase the claim in the main text as scoped to "the papers we examined," and treat a full systematic survey (with pre-registered search strings and dual coding, in the style of the tabular-QML audit of Bowles et al., 2024) as future work.

Paper	classical-network baseline	trained-circuit surrogate	reports χ / entanglemen
DisCoCat / `lambeq` [Lorenz et al., 2021]	–[a]	no	no
`QSANN` [Li et al., 2022]	yes	no	no
`QMSAN` [Chen et al., 2024]	yes	no	no
`CLAQS` [Chen et al., 2025]	yes	no	no

Table 4: Core reviewed set. [a]DisCoCat compares to a bag-of-words/word-sequence baseline rather than a matched network. None reports a classical surrogate of the trained circuit or a simulability diagnostic.

B Model circuits and glossary

All models are reimplemented in PennyLane; `lambeq` produces the DisCoCat circuits. We give each model's trained state below; the audit truncates that state (Algorithm 1).

`vqc_text` (reference). Bag-of-embeddings feature $x \in \mathbb{R}^n$ angle-encoded on n qubits, L layers of $R_y(x_i)$ (re-uploaded), trainable R_y, R_z, and a CNOT ring; readout $\langle Z_0 \rangle$ (Fig. 3).

`QSANN` (Gaussian-projected quantum self-attention). Per token, the same strongly-entangling template ($H^{\otimes n}$, then R_x/R_y layers with CNOT chains) prepares $|\psi_s\rangle = U_{enc}(x_s)H^{\otimes n}|0\rangle$; three trainable ansätze U_q, U_k, U_v give a query scalar $\langle Z_0 \rangle$, a key scalar, and a d-dimensional value vector of single-qubit Pauli expectations. Attention $\alpha_{s,j} = \exp(-(\langle Z_q \rangle_s - \langle Z_k \rangle_j)^2)$, the residual sum, mean-pool, and sigmoid are all classical. The audit therefore acts *per token* on the pure states $U_{q,k,v}|\psi_s\rangle$, and the classical head is the composition function.

`QMSAN` (mixed-state self-attention). Per token, a data-reuploading circuit ($R_x(x_i)$; $L\times$[IsingZZ ring, R_y layer]; re-upload) prepares a pure state whose reduced density matrix on the first $n/2$ qubits is the query/key object; attention is the Hilbert–Schmidt overlap $\alpha_{s,j} = \mathrm{tr}(\rho_{q,s}\sigma_{k,j})$. We audit this on the *purification*: we truncate the pure n-qubit state to bond dimension χ, then take its partial trace to form ρ, σ. Values are per-qubit $\langle Z_i \rangle$.

`CLAQS` (all-quantum token mixer). On q=8 data qubits, each token's ansatz-14 embedding unitary U_j is combined by a learnable, ℓ_1-normalised complex linear-combination-of-unitaries $M = \sum_j \bar{b}_j U_j$; a learnable QSVT polynomial $P_c(M) = \sum_{k=0}^{5} c_k M^k$ and a window feed-forward unitary U_{FF} produce $|\psi\rangle = \mathrm{normalise}(U_{\mathrm{FF}}\, P_c(M)\, |0^8\rangle)$; readout is 24 XYZ Pauli expectations into a two-layer MLP. Because $|\psi\rangle$ is a matrix polynomial rather than a gate sequence, the audit consumes the trained statevector directly. At q=8 the exact MPS bond dimension is 16, which is why this is the one model where the χ-axis has room to resolve structure.

`discocat`. A sentence's pregroup structure becomes one circuit: single-qubit word states (R_x / Euler $R_x R_z R_x$), IQP ansätze for multi-qubit words (per layer: H on all wires, then a chain of adjacent CR_z), a parameter-free GHZ for the relative pronoun, and grammar cups realised as Bell effects (post-select the paired qubits on $|0\rangle$). We audit the whole-sentence pure state before post-selection, then post-select and read the open wire's Born rule.

Glossary (for NLP readers). **NISQ**: noisy intermediate-scale quantum hardware. **PQC**: parameterised quantum circuit. **Ansatz**: a fixed circuit template whose rotation angles are trainable. R_x, R_y, R_z: single-qubit rotations; **CNOT**/CR_z: two-qubit entangling gates. **Statevector**: the 2^n complex amplitudes of an n-qubit pure state. **MPS** (matrix product state): a tensor-network factorisation whose **bond dimension** χ caps the entanglement it can represent; χ=1 is a product (unentangled) state. **Entanglement entropy** $\bar{S}$: bipartite von Neumann entropy of the state (nats). **IQP / LCU / QSVT**: instantaneous-quantum-polynomial ansatz / linear combination of unitaries / quantum singular-value transformation. **CSIM/QESIM**: classically simulable / classically simulable up to data loading.

C Datasets and preprocessing

MC (meaning classification). We use two variants. MC-REAL is the Lorenz et al. set (70 train / 30 dev / 30 test; 17-word vocabulary; 3–4 word cooking-vs-computing sentences), obtained from the authors' public resources repository; POS tags are stripped (`word_N`→`word`) since our bag-of-token models do not use them and the DisCoCat reader re-parses. MC-SYNTH is a self-contained stand-in generated from disjoint food/IT subject–verb–object templates (65 per class), used only for the controlled scaling study; being disjoint-vocabulary it is near-linearly-separable, which is why every model saturates on it.

RP (relative pronoun). The Lorenz et al. RELPRON-derived set (74 train / 31 test; 115-word vocabulary; four-token noun phrases), distinguishing subject- from object-relative clauses. We carve a 20% validation split from train. RP is the discriminating benchmark and, with only 74 training items, the fragile one—the original papers note near-degenerate behaviour here, which we reproduce.

SST-2. A balanced subset of the Stanford Sentiment Treebank binary task (600 train / 200 test, drawn from the train and labelled-validation splits respectively via the HuggingFace datasets server), sequence length capped at 16 tokens. A genuinely non-separable, real-text task; accuracies here are in the 0.5–0.65 range, not saturated.

Embeddings and tokenisation. For `qsann`, `qmsan`, `claqs`, and `vqc_text` we use a trainable per-token embedding (dimension equal to the model's feature width); `vqc_text` averages token embeddings (bag-of-embeddings), the attention models keep per-token vectors, and `claqs` maps each token embedding through a trainable linear layer to ansatz-14 angles. DisCoCat does not use embeddings: each word's parameters *are* its gate angles, trained via the `lambeq` pipeline with the offline `cups_reader` (the trained Bobcat grammar parser's model host is no longer reachable, an acknowledged fidelity cost on RP). Splits are seed-stable; the exact indices and the synthetic generator are released with the harness so every split is reproducible.

D Configurations, compute, and robustness

Hyperparameters. Table 5 lists the per-model settings. Quantum models train with Adam (AdamW for `claqs`); `discocat` trains with SPSA (a=0.05, c=0.06) as in the original pipeline. Quantum-model weights are initialised $\mathcal{N}(0, 0.01)$; all runs use seeds $\{1, 2, 3\}$. The matched classical baseline is logistic regression on bag-of-words counts. Quantum-model feature widths follow the source papers (n=2 qubits on MC, n=4 on RP; q=8 for `claqs`).

Model	qubits	depth/layers	optimiser	epochs
`vqc_text`	4–10	2	Adam (0.05)	30–40
`qsann`	2 / 4	D_{enc}=1, D_{qkv}=1	Adam (0.05)	40
`qmsan`	2 / 4	L=1 (IsingZZ ring)	Adam (0.05)	40
`claqs`	8	L=1, QSVT degree 5	AdamW (0.05)	10–12
`discocat`	3–7	IQP L=1	SPSA	120

Table 5: Per-model training configuration. Full configs and the exact commands are released with the harness.

Compute. All experiments run on a single Apple M1 CPU (8 GB RAM, no GPU). The audit is deterministic given a trained model—expectation values are computed exactly, not sampled—so the only stochasticity is training. Statevector simulation is memory-light at these sizes (a 16-qubit state is $\approx$ 1 MB); the binding constraint is training the attention models, not the audit. `claqs` is the exception: a faithful full-scale reproduction of its published 91.6% on SST-2 needs a GPU, so we run it at reduced scale (100 training examples, $\leq$ 6-token windows, 10 epochs); its SST-2 accuracy is consequently near chance and we report the audit, not an accuracy reproduction.

Robustness of $\chi^\star$ to the tolerance. Equation 1 defines $\chi^\star$ against a tolerance τ. We compute it under three definitions: τ_{gen} (the model's train–test generalization gap, our default), τ_{ci} (the half-width of a 95% bootstrap CI on full-χ accuracy), and τ_{agree} (smallest χ with prediction agreement $\pi(\chi) \geq 0.95$). Across all runs the three agree on $\chi^\star$ in the large majority of cases—χ=1 for the attention/mixer models and χ=2 for `discocat`—and where they differ they do so by a single bond-dimension step on borderline, low-accuracy RP runs. The headline conclusion is therefore not an artefact of the tolerance choice.